VARUN NATARAJ ALAGAPPAN

Software Development Engineer | Backend & Full-Stack

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WORK EXPERIENCE

Software Engineer 2 - Visteon Technical and Services Pvt Ltd | Jan 2024 - Present

- Architected and deployed **NexDiag to production** with Java 21, Spring Boot, Apache POI and XML parsing for CAPL/Robot generation from Excel/CDD requirements; adopted by **90% of automation/validation teams**; monitor production incidents and application logs.
- Built **automatic per-test-case CAN logging** in generated CAPL suites: each test writes its own .asc file with timestamped Tx/Rx frames and PASS/FAIL results; integrated REQ/RSP timestamps into an offline report tool.
- Added **Electron project persistence with embedded SQLite (sql.js/WebAssembly)**: autosaved DID tables, uploaded sheets and settings; maintained version history, restored the last project on startup, and enabled .nexdiag export/import with merge for multi-engineer collaboration.
- Implemented token-based authentication, persistent generated scripts and metadata, and per-test-case execution event logs in NexDiag.
- Led **Unified GUI** development with React, TypeScript, Electron and FastAPI; containerized services with Docker/Kubernetes, supported Linux deployments and investigated production incidents; achieved **95% team adoption and 90% less manual effort**.
- Built **REST API backends** for restbus/CAN handlers used by Robot Framework, Unified GUI and diagnostic tools; integrated FastAPI, WebSockets and RabbitMQ for communication and diagnostic workflows.
- Built a **Rust CAN Log Parser** that processed a **2.10 GB / 19.81-million-line log in approximately 3 seconds**, with SQLite indexing and reports; optimized Python automation and V-RoboBus workflows.
- Architected **RoboFIT/Robot Framework** and reusable Python libraries for hardware/software interaction, integrating Selenium and Appium; reduced manual effort by **95%**, with adoption by approximately **95% of Visteon automation teams globally**.
- Automated **Jenkins build, test and deployment pipelines** for scheduled hardware flashing, remote-PC execution and emailed reports; reduced manual effort by **90%** and rolled out across Visteon.
- Developed **JIRA REST API integrations** to capture failures, create defects and rerun linked tests on target builds; built a local LLM workflow that generated test cases from requirements in JIRA-standard formats.
- Built Android-to-DUT Bluetooth/Wi-Fi automation with Python, Robot Framework, Appium and ADB, including automatic log capture; applied data structures and algorithms to optimize runtime and cut manual effort by **90%**.
- Maintained pytest unit tests and Playwright end-to-end coverage for authentication, script generation and downloads; participated in code reviews, CI and Agile delivery, and coordinated customer issue resolution with development, DevOps and SRE teams.

TECHNICAL SKILLS

Languages & Backend: Python, Java, Rust, C++, JavaScript/TypeScript, SQL; Spring Boot, FastAPI, Node.js, REST APIs, WebSockets, RabbitMQ; data structures and algorithms

Frontend & Data: React, Electron, Kotlin, Jetpack Compose; MySQL, PostgreSQL, SQLite; JWT, token-based authentication, RBAC

Delivery & Quality: Linux, shell scripting, Git, Jenkins, GitHub Actions, Docker, Kubernetes, AWS; pytest, JUnit 5, Mockito, Testcontainers, Playwright, k6, Robot Framework; debugging and root-cause analysis

PROJECTS

LoanLens - Fintech Loan Risk Analyzer | Java Spring Boot, React, MySQL | Dec 2025 - Mar 2026

- Built nine JWT-secured APIs; validated p95 <800 ms at 100 concurrent users with k6 and tested with JUnit 5, Mockito, Testcontainers and Playwright; deployed through Docker, AWS EKS and GitHub Actions.

Best & Banting - Medical Equipment E-Commerce | React, Node.js, PostgreSQL | Dec 2025

- Secured Express APIs with JWT, bcrypt, rate limiting, RBAC and parameterized SQL; built 3D product views, analytics and exports. Live demo

ScoreStep - Credit Health Coach | Kotlin, Jetpack Compose | Jun 2026

- Built an offline MVVM Android app with StateFlow, Room and Hilt for credit-health plans, DTI/eligibility checks and EMI simulations.

Fleet Management Solution with RBAC | C++, SQL | Jun 2025

- Implemented SHA256 authentication, role-based access, rental/return workflows and SQL reports; validated 500+ login attempts and database audit tracking.

Virtual Automotive Control System - Gear Up Hackathon | C | Mar 2025

- Simulated engine, transmission, braking, HVAC and cluster systems; won Best Team among 100+ teams at Visteon & KICIT.

Retinal Disease Diagnosis using Deep Learning | May 2024

- Built a Multi Scale-Attention model for fundus/OCT images with approximately 93% accuracy; served predictions through Django.

ACHIEVEMENTS

- **HackerRank Problem Solving:** 301 challenges solved | Rank 1,070 | 16,605 points (26 Oct 2026).
- **Visteon Silver Award** for contributions to Visteon's growth; **Best Team, Gear Up Hackathon** among 100+ teams.

EDUCATION

B.E. Electronics and Communication Engineering - SSN College of Engineering, Chennai | May 2024
CGPA: 8.6/10